Horizon Capital

User Guide

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Horizon Capital is an autonomous AI hedge fund platform that runs entirely in your terminal. It employs a multi-agent architecture -- strategy analysts, risk managers, compliance officers, and an AI CIO -- debating and collaborating to generate, validate, and execute trading signals across Indian, US, and crypto markets. This guide walks you through every feature, from first launch to advanced CIO governance.

1. User Personas

Horizon Capital serves different roles. Find yours below to understand which commands and workflows matter most to you.

Paper Trader

Who you are: You are learning the platform, testing strategies, or evaluating Horizon Capital before committing real capital. Everything runs in simulation -- no broker connections needed.

Your daily workflow:

- Launch the platform: python -m app
- Check agent status: status
- Watch incoming signals: signals
- Review portfolio NAV: portfolio
- Monitor data feeds: data
- Explore the agent office: office

Key commands: status, signals, portfolio, data, symbols, office

Quantitative Analyst

Who you are: You focus on backtesting strategies, tuning models, and analyzing performance attribution. You want to know which strategies are generating alpha and why.

Your daily workflow:

- Run backtests on target symbols: backtest AAPL.NASDAQ --days 90
- Compare strategy results: backtest results
- Review model registry: models
- Check Thompson Sampling rotation: model stats
- Analyze performance metrics: performance 30d
- Drill into attribution: attribution AAPL.NASDAQ

Key commands: backtest, performance, models, model stats, attribution, backtest results

Risk Manager

Who you are: You monitor Value-at-Risk, run stress tests against historical crises, and have the authority to halt trading via the kill switch.

Your daily workflow:

- Check risk dashboard: risk
- Review market regime: regime
- Run stress tests: stress-test
- Monitor alerts: alerts
- Review alert rules: alert rules
- Emergency halt if needed: kill-switch

Key commands: risk, stress-test, kill-switch, alerts, alert rules, regime

CIO (Human Mode)

Who you are: You are the Chief Investment Officer operating in HUMAN mode. You approve or reject agent recommendations, override regime classifications, and adjust risk tolerance in real time.

Your daily workflow:

- Review agent debates: debate
- Enter CIO chat for natural language control: cio
- Override market regime if needed: override regime INDIA BULL
- Adjust risk tolerance: risk-tolerance 5.0
- Interrupt misbehaving agents: interrupt <agent_id>
- Generate daily report: report daily

Key commands: cio, override regime, risk-tolerance, interrupt, debate, toggle sentiment

Fund Manager (AI Mode)

Who you are: You have set CIO_MODE=AI and the system runs fully autonomously. You monitor performance and generate reports but do not intervene in trading decisions.

Your daily workflow:

- Check system health: health
- Review overall status: status
- Monitor performance: performance YTD
- Generate reports: report weekly
- Export for stakeholders: report export csv

Key commands: status, performance, report, health, workers

2. Getting Started

Prerequisites

- Python 3.11 or newer (tested up to Python 3.14)
- pip (Python package manager)
- 4 GB RAM minimum (8 GB recommended)
- Internet connection for market data feeds

Installation

```
# Clone or download the repository
cd horizon-capital/backend

# Create a virtual environment (recommended)
python -m venv .venv
source .venv/bin/activate    # Linux/macOS
.venv\Scripts\activate       # Windows

# Install dependencies
pip install -e .
```

First Launch

```
python -m app
```

On first run, the onboarding wizard will guide you through initial setup.

Onboarding Wizard

The wizard presents two paths:

Quick Start (recommended for beginners)

- Sets PAPER_TRADING mode
- Uses LOCAL storage (SQLite, no Docker needed)
- Configures default trading universe (NIFTY 50, S&P 500, top crypto)
- Skips API key configuration (uses free Yahoo Finance data)
- You can start exploring immediately

Advanced Setup

- Choose your system mode (Paper Trading / Human Approval / Fully Autonomous)
- Configure LLM API keys (Claude, GPT, Gemini)
- Set up broker connections (Alpaca, Zerodha)
- Customize trading universe
- Configure alert channels (Telegram, email)
- Set risk parameters

You can re-run the wizard at any time with setup from the REPL or python -m app --setup from the command line.

3. Platform Modes

Trading Modes

Mode	Description	Who Decides
PAPER_TRADING	Simulated trades, no real money	Agents auto-execute (paper)
HUMAN_APPROVAL	Agents propose, human approves	CIO (you) approves each trade
FULLY_AUTONOMOUS	Agents trade without approval	AI CIO governs

Switch modes at runtime:

```
mode PAPER_TRADING
mode HUMAN_APPROVAL
mode FULLY_AUTONOMOUS
```

CIO Mode

Mode	Behavior
HUMAN	You act as CIO -- approve trades, override regimes, adjust risk
AI	AI CIO agent makes all governance decisions autonomously

Set via environment variable CIO_MODE=HUMAN or CIO_MODE=AI in your .env file.

Storage Backends

Backend	Database	Message Bus	Requirements
LOCAL (default)	SQLite	In-memory bus	Nothing -- works out of the box
POSTGRES	PostgreSQL	Redis pub/sub	Docker or managed services

LOCAL mode is recommended for individual use. POSTGRES mode is for multi-instance deployments.

4. Step-by-Step Workflows

Workflow 1: First Launch and Paper Trading

Goal: Get the platform running and see agents in action.

```
Step 1:  python -m app
         -> Onboarding wizard runs (choose Quick Start)

Step 2:  status
         -> See all agents grouped by department:
            Front Office:  Strategy, Regime, Sentiment agents
            Middle Office: Risk, Compliance, VaR agents
            Back Office:   Operations, Accounting, IT agents
            Executive:     CIO agent

Step 3:  data
         -> See live market data, OHLCV counts, feed health

Step 4:  signals
         -> Watch trading signals as agents generate them

Step 5:  portfolio
         -> Check NAV, cash balance, and positions

Step 6:  office
         -> Watch the animated ASCII art agent office
```

Workflow 2: Monitoring Market Data

Goal: Understand what data the platform is ingesting.

```
Step 1:  symbols
         -> View trading universe:
            India:  RELIANCE.NSE, TCS.NSE, INFY.NSE, ...
            US:     AAPL.NASDAQ, MSFT.NASDAQ, GOOGL.NASDAQ, ...
            Crypto: BTC-USD, ETH-USD, SOL-USD, ...

Step 2:  data
         -> View latest prices, OHLCV bar counts,
            feature cache status, forex rates (INR/USD),
            feed health, circuit breaker states

Step 3:  workers
         -> Check background worker status:
            Data Ingestion:        polling Yahoo/CoinGecko/Forex
            Feature Computation:   23 technical indicators
            Performance Analytics: Sharpe, alpha calculations
```

Workflow 3: Analyzing Signals and Debates

Goal: Understand why agents are recommending specific trades.

```
Step 1:  signals
         -> See recent signals with direction (BUY/SELL/HOLD)
            and strength (0.0 to 1.0)

Step 2:  debate
         -> Read agent debate summaries:
            Strategy Agent argues for momentum entry
            Risk Agent raises VaR concerns
            Compliance Agent checks position limits
            CIO Agent makes final decision

Step 3:  regime
         -> See market regime per scope:
            INDIA:  BULL (confidence: 0.82)
            US:     NEUTRAL (confidence: 0.65)
            CRYPTO: HIGH_VOL (confidence: 0.91)
```

Workflow 4: Running a Backtest

Goal: Test a strategy on historical data.

```
Step 1:  backtest AAPL.NASDAQ --days 90
         -> Runs walk-forward backtest on AAPL with 90 days of data
            Tests all strategies: momentum, mean_reversion,
            volatility, crisis

Step 2:  backtest AAPL.NASDAQ --days 252 --strategy momentum
         -> Run a specific strategy over 1 year

Step 3:  backtest results
         -> View comprehensive results:
            Total Return, Sharpe Ratio, Max Drawdown,
            Win Rate, Profit Factor, Calmar Ratio
```

Workflow 5: Managing Risk

Goal: Monitor and control portfolio risk.

```
Step 1:  risk
         -> View risk dashboard:
            VaR 95%:        -2.3%
            VaR 99%:        -3.8%
            Max Drawdown:   -5.1%
            Current DD:     -1.2%
            Kill Switch:    INACTIVE

Step 2:  stress-test
         -> Run all historical scenarios:
            GFC_2008, COVID_2020, TAPER_TANTRUM_2013,
            FLASH_CRASH_2010, CRYPTO_WINTER_2022

Step 3:  stress-test COVID_2020
         -> Detailed impact analysis for a specific scenario

Step 4:  risk-tolerance 3.0
         -> Tighten max drawdown to 3% (range: 1.0-50.0%)

Step 5:  kill-switch
         -> ACTIVATE to halt all trading immediately
         -> DEACTIVATE to resume trading
```

Workflow 6: Generating Reports

Goal: Create portfolio reports for review or export.

```
Step 1:  report daily
         -> Generate daily portfolio report:
            P&L summary, top winners/losers,
            risk metrics, signal summary

Step 2:  report weekly
         -> Weekly aggregate report

Step 3:  report monthly
         -> Monthly performance review

Step 4:  report export csv
         -> Export last report as CSV file

Step 5:  report export json
         -> Export last report as JSON
```

Workflow 7: Using CIO Chat Mode

Goal: Interact with the system using natural language.

```
Step 1:  cio
         -> Enter CIO chat mode

Step 2:  Type natural language queries:
         "What is our current risk exposure?"
         "Why did the strategy agent recommend buying AAPL?"
         "Show me the top performing strategies this month"
         "Reduce exposure to crypto markets"

Step 3:  Type 'exit' to leave CIO chat mode
```

5. Command Reference

System Commands

Command	Description	Example
help	Show all available commands	help
status	Show all agent statuses by department	status
departments	Department-level overview	departments
workers	Background worker status	workers
config	Show current configuration	config
setup	Re-run onboarding wizard	setup
office	Show animated agent office (ASCII art)	office
quit / exit	Shutdown Horizon Capital	quit

Portfolio and Trading

Command	Description	Example
portfolio	Current portfolio (NAV, cash, positions)	portfolio

Command	Description	Example
signals	Recent trading signals	signals
exposures	Current exposures by symbol	exposures
symbols	Trading universe (India, US, Crypto)	symbols
data	Live market data and feed health	data

Risk Management

Command	Description	Example
risk	Risk dashboard (VaR, drawdown, kill switch)	risk
regime	Market regime per scope with confidence	regime
stress-test	Run all crisis scenarios	stress-test
stress-test <SCENARIO>	Run specific scenario	stress-test COVID_2020
kill-switch	Toggle emergency trading halt	kill-switch
risk-tolerance <N>	Set max drawdown % (1.0-50.0)	risk-tolerance 5.0

Strategy and Analysis

Command	Description	Example
backtest <SYMBOL>	Run walk-forward backtest	backtest AAPL.NASDAQ --days 90
backtest results	Show last backtest results	backtest results
debate	Recent agent debate summaries	debate
performance [period]	Performance metrics (1d/7d/30d/90d/YTD/ALL)	performance 30d
attribution [symbol]	Per-trade attribution breakdown	attribution AAPL.NASDAQ

Models and ML

Command	Description	Example
models	Registered ML models with status and Sharpe	models
model stats	Thompson Sampling rotation statistics	model stats

Reports and Alerts

Command	Description	Example
report [period]	Generate report (daily/weekly/monthly)	report daily
report export [fmt]	Export report (csv/json)	report export csv
alerts [severity]	Show alerts (INFO/WARNING/CRITICAL/EMERGENCY)	alerts CRITICAL
alert rules	Show alert rules and channel status	alert rules
health	System health dashboard	health

CIO and Governance

Command	Description	Example
cio	Enter CIO natural language chat	cio

Command	Description	Example
override regime <SCOPE> <REGIME>	Manually set market regime	override regime INDIA BULL
interrupt <agent_id>	Pause a specific agent	interrupt strategy_agent
toggle sentiment	Enable/disable sentiment agent	toggle sentiment
mode <MODE>	Switch trading mode	mode HUMAN_APPROVAL
agents config	Show agent LLM configuration	agents config

6. Configuration Guide

API Keys

Configure in your .env file in the backend directory:

```
# LLM Providers (optional -- agents work with any one provider)
ANTHROPIC_API_KEY=sk-ant-...     # Claude (recommended)
OPENAI_API_KEY=sk-...            # GPT-4
GOOGLE_API_KEY=AI...             # Gemini

# Brokers (optional -- not needed for paper trading)
ALPACA_API_KEY=PK...
ALPACA_API_SECRET=...
ZERODHA_API_KEY=...
ZERODHA_API_SECRET=...

# Data Providers (optional -- Yahoo Finance is free and default)
POLYGON_API_KEY=...
```

Risk Parameters

Adjustable at runtime or via .env:

```
# Maximum portfolio drawdown before kill switch activates
MAX_DRAWDOWN_PCT=5.0

# Position sizing limits
MAX_POSITION_PCT=10.0        # Max % of NAV in a single position
MAX_SECTOR_PCT=30.0          # Max % of NAV in a single sector

# VaR confidence levels
VAR_CONFIDENCE_95=0.95
VAR_CONFIDENCE_99=0.99
```

Trading Universe

The default universe includes:

- India: RELIANCE.NSE, TCS.NSE, INFY.NSE, HDFCBANK.NSE, and more
- US: AAPL.NASDAQ, MSFT.NASDAQ, GOOGL.NASDAQ, AMZN.NASDAQ, and more
- Crypto: BTC-USD, ETH-USD, SOL-USD, and more

Customize via the onboarding wizard (setup) or by editing the configuration.

Alert Channels

```
# Telegram notifications
TELEGRAM_BOT_TOKEN=...
TELEGRAM_CHAT_ID=...

# Email notifications
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=...
SMTP_PASSWORD=...
ALERT_EMAIL_TO=you@example.com
```

7. Troubleshooting

Platform will not start

Symptom: Error on python -m app

- Ensure you are in the backend/ directory
- Check Python version: python --version (must be 3.11+)
- Install dependencies: pip install -e .
- Check for port conflicts if using POSTGRES mode

No market data appearing

Symptom: data command shows empty or stale values

- Check internet connection
- Run workers to verify Data Ingestion worker is running
- Yahoo Finance has rate limits -- wait a few minutes and retry
- Check circuit breaker states in data output (CLOSED = healthy)

Agents stuck or not responding

Symptom: status shows agents in ERROR state

- Run workers to check if orchestrator is running
- Workers auto-restart on crash (check logs for details)
- Use interrupt <agent_id> then wait for auto-restart
- As a last resort, restart the platform

Kill switch activated unexpectedly

Symptom: Trading halted, risk shows kill switch ACTIVE

- Check risk for current drawdown vs max drawdown threshold
- Use risk-tolerance <N> to adjust if threshold is too tight
- Use kill-switch to deactivate when ready to resume

Backtest returns no results

Symptom: backtest results shows nothing

- Ensure you ran a backtest first: backtest AAPL.NASDAQ --days 90
- Check that market data exists for the symbol: data
- Try a shorter period: --days 30

LLM agents not working

Symptom: Agents produce generic or empty responses

- Verify API key is set: config (keys are masked but shown)
- At least one LLM provider key is needed for full agent functionality
- The platform works without LLM keys but with reduced agent intelligence
- Re-run setup to configure API keys

High memory usage

Symptom: Platform consuming excessive RAM

- LOCAL mode with SQLite is the most efficient
- Reduce trading universe size via setup
- Close other applications
- Consider running in headless mode: python -m app --headless

Common error messages

Error	Cause	Fix
ModuleNotFoundError	Missing dependency	pip install -e .
Address already in use	Port conflict (POSTGRES mode)	Stop other services or change port
Connection refused	Redis/Postgres not running	Start Docker services or use LOCAL mode
Rate limit exceeded	Too many API calls	Wait and retry; circuit breaker will handle it
Invalid API key	Wrong or expired key	Check .env file, re-run setup